

MANUAL HYPERPARAMETER SELECTION FORM

2D CNN Model — Dogs vs. Cats Image Classification

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Use this form to record, justify, and approve each hyperparameter value chosen by hand before a training run. The "Notebook Default" column shows the value currently used in `Loading_Dataset_and_CNN.ipynb` for reference — enter the value actually used for this run under "Chosen Value", even if it matches the default.

1. Project / Experiment Information

Project / Experiment Title	CNN Activity w/ hyperparameters tuning
Model Architecture	2D Convolutional Neural Network (Conv2D)
Dataset Used	Kaggle Dogs vs. Cats images (Dog/ and Cat/ folders)
Notebook / File Reference	Loading_Dataset_and_CNN.ipynb
Prepared By	Bernard G. Yasay
Date	October 6, 2026
Experiment / Trial No.	30 Trials

2. Data Preprocessing Parameters

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification
Image Size (IMG_SIZE, square)	<i>e.g. 50 / 80 / 100 / 128 / 150</i>	100	100	The Image Size (IMG_SIZE) was set to 100 × 100 pixels to provide a practical balance between image detail, computational efficiency, and model complexity.
Color Mode	<i>Grayscale / RGB</i>	Grayscale	Grayscale	The grayscale color mode was selected because the classification task primarily depends on the shape, texture, edges, and intensity patterns of the objects rather than their color information.
Pixel Normalization	<i>X/255.0 / none</i>	<i>X/255.0</i>	X / 255.0	Normalizing the pixel values prevents large input values from dominating the learning process and provides the CNN with inputs on a consistent numerical scale.

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification
Shuffle Training Data	Yes / No	Yes (<i>random.shuffle</i>)	Yes	Shuffling helps prevent the model from learning patterns that may be associated with the original ordering of the dataset rather than the actual characteristics of the image classes.
Held-out Test Set (separate from training)	Yes / No	No (<i>validation_split only</i>)	No.	It only splits the dataset into training and validation sets (X_train, X_val, y_train, y_val) using test_size=0.2.
Validation Split (of training set)	0.0 - 0.3	0.1	0.2	The validation split was set to 0.2 (20%) to provide a sufficient portion of the dataset for evaluating the CNN's performance on images that are not used for training.

3. Model Architecture Hyperparameters

Trial 1

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Filters	16 / 32 / 64 / 128	64	64	64 filters is used to provide the CNN with sufficient capacity to extract a diverse set of low-level visual features from the input images.	0.9896	0.8726
Conv2D Layer 1 – Kernel Size	(3,3) / (5,5)	(3,3)	(3,3)	(3,3) provides an effective balance between capturing local image features and maintaining computational efficiency.		

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Activation	<i>relu / tanh / elu / leaky_relu</i>	<i>relu</i>	relu	The ReLU (Rectified Linear Unit) activation function was used in Conv2D Layer 1 to introduce non-linearity into the CNN, enabling the model to learn more complex visual patterns from the input images.		
Pooling Layer 1 – Type & Pool Size	<i>MaxPooling2D / AveragePooling2D; (2,2) / (3,3)</i>	<i>MaxPooling2D, (2,2)</i>	<i>MaxPooling2D, (2,2)</i>	The MaxPooling2D layer with a (2,2) pool size was used to reduce the spatial dimensions of the feature maps produced by the convolutional layer while retaining the most significant features.		
Dense (Fully Connected) Units	<i>16 / 32 / 64 / 128</i>	<i>64</i>	64	The Dense (Fully Connected) layer was set to 64 units to provide sufficient capacity to combine and interpret the features extracted by the preceding convolutional and pooling layers.		
Dropout Rate	<i>0.0 - 0.5</i>	<i>None (not used)</i>	<i>0.0 / 0.1 / 0.2 / 0.3 / 0.4 / 0.5</i>	The Dropout Rate is not one fixed value in the optimized model. Options / Range Considered: 0.0 to 0.5, in increments of 0.1		
Output Layer Activation	<i>sigmoid (binary) / softmax (multi-class)</i>	<i>sigmoid</i>	<i>sigmoid</i>	The sigmoid activation function was used in the output layer because the CNN performs a binary classification task, where the model must distinguish between two classes.		

4. Training Hyperparameters

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Loss Function	<i>binary_crossentropy / categorical_crossentropy</i>	<i>binary_crossentropy</i>	<i>binary_crossentropy</i>	The binary cross-entropy loss function was used because the CNN is designed for a binary classification problem, where each input image belongs to one of two possible classes.		
Optimizer	<i>adam / sgd / rmsprop</i>	<i>adam</i>	<i>adam</i>	The Adam (Adaptive Moment Estimation) optimizer was used because it provides an efficient and adaptive method for updating the CNN's weights during training.		
Learning Rate	<i>0.1 / 0.01 / 0.001 / 0.0001</i>	<i>0.001 (Adam default)</i>	0.0001 to 0.01	The Learning Rate is not a single fixed value because it is hyperparameter-tuned.		
Batch Size	<i>8 / 16 / 32 / 64 / 128</i>	32	128	With a batch size of 128, the CNN processes 128 training images before updating its weights, allowing the model to utilize the available computational resources efficiently while reducing the number of weight updates required in each epoch.		

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Number of Epochs	<i>e.g. 10 - 200</i>	<i>200</i>	50	The number of epochs was set to 50 to provide the CNN with sufficient opportunities to learn and refine the important features present in the training images.		
Early Stopping	<i>Yes (patience = ____) / No</i>	<i>No</i>	Yes	Early Stopping was used to prevent the CNN from overfitting and to avoid unnecessary training once the model stopped improving on the validation data.		

5. Evaluation Results (after training with chosen values) - Final Model

Metric	Value Obtained	Target / Acceptable Range
Training Accuracy	98.96%	$\geq 90\%$ target
Validation Accuracy	87.26%	$\geq 90\%$ target
Test Accuracy	N/A	Does not have a separate held-out test set.
Test Loss	N/A	Does not have a separate held-out test set.
Precision / Recall / F1 (macro avg)	87.27% / 87.26% / 87.26%	Preferably ≥ 0.90 macro F1

5. a. Paste here all screenshots/figures of the tuned model

Model: "sequential_1"

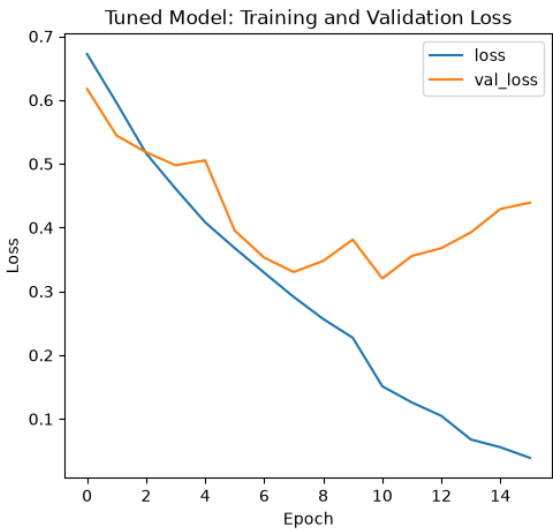
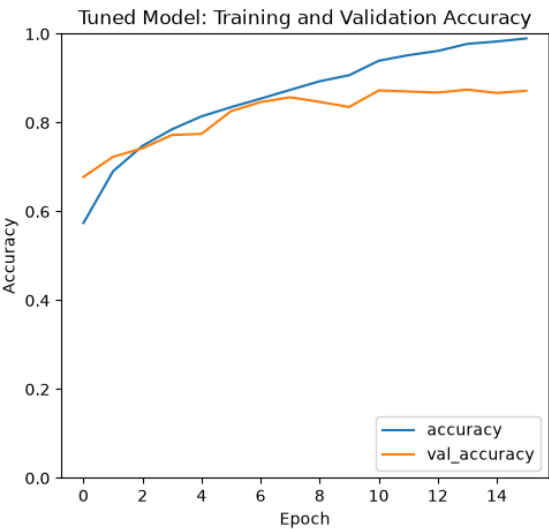
Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 100, 100, 96)	2,496
activation_6 (Activation)	(None, 100, 100, 96)	0
max_pooling2d_3 (MaxPooling2D)	(None, 50, 50, 96)	0
conv2d_4 (Conv2D)	(None, 50, 50, 64)	153,664
activation_7 (Activation)	(None, 50, 50, 64)	0
max_pooling2d_4 (MaxPooling2D)	(None, 25, 25, 64)	0
conv2d_5 (Conv2D)	(None, 25, 25, 32)	51,232
activation_8 (Activation)	(None, 25, 25, 32)	0
max_pooling2d_5 (MaxPooling2D)	(None, 12, 12, 32)	0
conv2d_6 (Conv2D)	(None, 12, 12, 192)	55,488
activation_9 (Activation)	(None, 12, 12, 192)	0
max_pooling2d_6 (MaxPooling2D)	(None, 6, 6, 192)	0
flatten_1 (Flatten)	(None, 6912)	0
dense_3 (Dense)	(None, 128)	884,864
activation_10 (Activation)	(None, 128)	0
dropout_1 (Dropout)	(None, 128)	0
dense_4 (Dense)	(None, 1)	129
activation_11 (Activation)	(None, 1)	0

Total params: 2,295,748 (8.76 MB)

Trainable params: 1,147,873 (4.38 MB)

Non-trainable params: 0 (0.00 B)

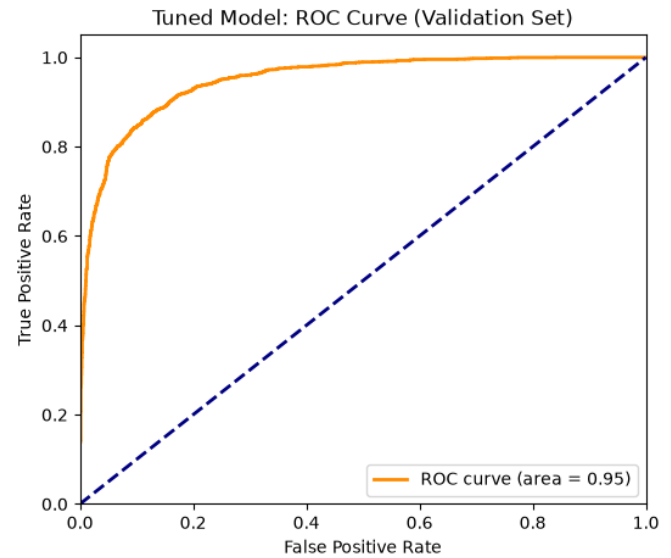
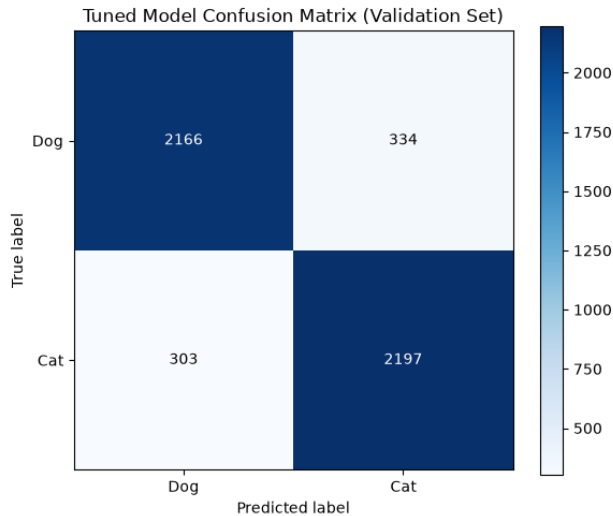
Optimizer params: 1,147,875 (4.38 MB)



157/157 16s 98ms/step

Tuned Model Confusion Matrix (validation set):

```
[[2166 334]
 [ 303 2197]]
```



6. Overall Justification / Remarks

Summarize why this combination of hyperparameters was chosen (e.g., prior trial results, literature basis, computational constraints) and any observations (overfitting, underfitting, convergence issues).

The chosen combination of hyperparameters was selected to achieve a balance between classification performance, computational efficiency, and model generalization. The combination of 100×100 grayscale images with pixel normalization was chosen to reduce the computational demand while capturing the essential structure and intensity of the image for classification. The CNN structure, with 64 filters, a 3×3 kernel, ReLU activation, 2×2 max pooling, and 64 dense units, was sufficient to capture the relevant image characteristics without being unnecessarily complex. For a binary classification problem, the sigmoid activation function and binary cross-entropy were appropriate, while the Adam optimizer was a logical choice to achieve efficient convergence. A batch size of 128 was sufficient for stable and efficient training, and the model was trained for up to 50 epochs with Early Stopping to ensure that the model would not overfit the data.

Based on the results in the notebook, I can conclude that the model learned the training set reasonably well, as evidenced by the high training accuracy ($\sim 98.96\%$) and lower validation accuracy ($\sim 87.26\%$). The significant gap between training and validation accuracy (around 11.70 percentage points) suggests that the model may be overfitting the training data to some degree. In this case, Early Stopping is crucial to prevent further overfitting and ensure that the validation performance is not compromised. The combination of hyperparameters that were selected is a reasonable choice for this task, as it provides a good balance between model capacity, training efficiency, and generalization performance. Nevertheless, the significant gap between training and validation accuracy indicates that further refinement of the hyperparameters or the addition of regularization techniques could be beneficial to improve the generalization performance of the model. The notebook does not evaluate the performance of the model on a held-out test set, so any conclusions should be based on the validation performance rather than test accuracy if a separate test set is available.